Bhavana Nare
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PROFESSIONAL SUMMARY

- Applied AI & ML Engineering Leader with 10+ years of experience architecting and deploying production-scale intelligent systems across cloud-native environments.
- Proven track record building and scaling LLM-driven platforms, computer vision systems, and hybrid ML-rule architectures that deliver measurable operational impact across enterprise ecosystems.
- Experienced in end-to-end ML lifecycle ownership—from experimentation and feature engineering to model deployment, evaluation, observability, and governance.
- Technical leader with hands-on expertise in distributed systems, RAG architectures, CI/CD automation, and MLOps best practices, driving reliable, auditable, and scalable AI platforms aligned with business objectives.

EDUCATION & RESEARCH

- **Master of Computer Science (Thesis)** — University of Georgia, Athens, GA, Aug 2021 – May 2023 | GPA: 3.7 / 4.0
- **Bachelor of Technology in Computer Science** — Sree Vidyanikethan Engineering College, Tirupati, India, Oct 2010 – Apr 2014 | GPA: 7.9 / 10

PROFESSIONAL EXPERIENCE

AI Engineer ***Rivian Automotive (May 2025 –Present)***

- Designed and deployed LLM-based AI systems using Vertex AI and RAG architectures to analyze 400–600 daily merge requests and generate contextual security recommendations.
- Built data preprocessing and feature engineering pipelines in Python to extract structured signals from source code, commit history, and security policies for model inference.
- Implemented GenAI guardrails using Vertex AI evaluation and prompt tuning workflows to reduce hallucinations and enforce policy-aware outputs.
- Improved vulnerability triage accuracy by reducing false positives by ~65% through structured evaluation datasets and feedback-driven model tuning.
- Developed Python ML services and REST APIs integrating Vertex AI models into GitLab CI/CD pipelines for automated merge-request analysis.
- Established model monitoring and evaluation pipelines tracking LLM output quality, response confidence, and system reliability.
- Collaborated with DevOps and platform teams to operationalize enterprise-scale GenAI systems across engineering workflows.

MLOps Engineer ***Robert Bosch (Aug 2023 –April 2025)***

- Designed and implemented end-to-end ML pipelines on Microsoft Azure to analyze telemetry data from 80+ repositories and predict integration risk patterns across CI/CD workflows.
- Built supervised machine learning models using Random Forest and scikit-learn in Python to classify pull-request instability based on engineered features such as defect density, change frequency, integration latency, and volatility metrics.
- Developed data preprocessing and feature engineering pipelines to transform raw repository telemetry into structured training datasets for ML models.
- Deployed ML models as containerized services and REST APIs integrated with Azure-based pipelines enabling scalable inference within internal automation platforms.
- Implemented model evaluation and monitoring frameworks tracking metrics such as precision, recall, and F1 score to ensure prediction reliability.
- Automated ML training and deployment using CI/CD pipelines (GitLab CI, Jenkins) enables reproducible model experiments and controlled releases.
- Built analytics dashboards to monitor prediction trends, model performance, and operational metrics for engineering teams.
- Mentored engineers on ML pipeline design, feature engineering, and MLOps practices.

Senior Software Engineer –Computer Vision ***Continental Automotive (May 2019 – July 2021)***

- Architected and delivered AUTOSIM, reducing manual validation effort by 60–70% and improving release confidence across perception teams.
- Served as Scrum Master for the COD component, facilitating sprint planning, backlog refinement, cross-team alignment, and stakeholder reviews.
- Contributed to ML-based object detection pipelines using YOLO architectures; supported iterative training, validation, and production-readiness evaluation.
- Led transition from 2D to 3D bounding box validation workflows, improving spatial accuracy and downstream decision reliability.
- Designed performance evaluation pipelines measuring precision, recall, mAP, IoU, and regression trends across releases.
- Mentored new engineers on perception of validation frameworks and ML evaluation methodologies.
- Collaborated with calibration, system safety, and integration teams to align ML outputs with safety-critical ADAS requirements.

ADDITIONAL EXPERIENCE

- **Python Developer & Data Analyst** — Teradata India Pvt. Ltd August 2018 – May 2019
- **Senior Software Engineer** — Tata Consultancy Services (TCS) June 2014 – August 2018

TECHNICAL SKILLS

- **Programming & Languages:** Python, Java, C++, SQL, Bash/Shell, JavaScript (React)
- **AI / ML & LLM Platforms:** Vertex AI, AWS Bedrock, PyTorch, scikit-learn, Pandas, NumPy, OpenCV
- **Cloud & MLOps:** AWS (Lambda, SageMaker, S3, DynamoDB, RDS, API Gateway, Step Functions), Google Vertex AI, Docker, Terraform, GitHub Actions, GitLab CI, Jenkins
- **Backend Frameworks:** Django, FastAPI, Flask
- **Data & Storage:** DynamoDB, PostgreSQL, Amazon RDS, Databricks, S3, Vector Databases (PG Vector)
- **Observability & Analytics:** CloudWatch, Structured Logging, Metrics & Alerting, GraphQL, Matplotlib, Hex
- **Data Storage & Retrieval:** DynamoDB, PostgreSQL, Amazon RDS, S3, pgvector (Vector Databases)
- **Engineering Practices:** Microservices Architecture, System Design, Performance Optimization, Agile Development

PUBLICATION

A Novel Computational Framework of Robot Trust for Human-Robot Teams - IEEE,2025

- Developed a computational trust model enabling autonomous agents to quantify and dynamically update trust in human teammates.
- Implemented a multi-agent framework integrating psychological trust factors into adaptive robot decision-making.

Interpretable Deep Learning for Biological Age Prediction: A Counterfactual Approach to Personalized Health Insights
Atlantis Press, 2026

- Designed an Explainable Time Series Regression (XTSR) framework integrating deep learning with counterfactual reasoning.
- Trained hybrid time-series models on large-scale wearable sensor data to generate interpretable biological age predictions.

CERTIFICATION

- **ISTQB Certified Tester (CTFL) -** ITB, *Dec 2020*
- **Build a Computer Vision App with Azure Cognitive Service –** Microsoft, Feb 2026
- **Dataiku Advanced Designer** — Dataiku, Mar 2026
- **Dataiku Core Designer** — Dataiku, Mar 2026
- **Dataiku ML Practitioner** — Dataiku, Mar 2026